

SEOKCHAE YOON

Ph.D. Candidate in Management Engineering (Information Systems Track), KAIST College of Business

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OVERVIEW

My research asks whether the digital and AI services that observe and act on individuals' finances and health genuinely serve those individuals, and what model and deployment design this requires. My dissertation pursues this question in healthcare AI: it builds a noise-robust smartphone-based depression screening model (job market paper, under review at *MIS Quarterly*), causally evaluates a digital therapeutic through randomized trials, and designs hybrid AI underwriting that balances risk protection with access. A second stream, anchored by my publication in *Information Systems Research*, examines how platform services reshape the economic lives of workers, from on-demand wage access to algorithmic matching in ride-hailing. Across both streams, I build AI models when the question requires one and verify effects with causal inference.

Research Interests: Healthcare IT · Trustworthy AI and Algorithmic Decision-Making · Platform Economics

Research Methods: Deep Learning · Causal Inference · Econometrics · Bayesian Statistics and Machine Learning

JOB MARKET PAPER

“Screening Mental Health with Smartphones: Understanding Real-World Noise in Smartphone-Generated Physiological Data”

Yoon, S., Oh, J., & Oh, W.

Under review, *MIS Quarterly*.

To be presented at the 2026 Conference on Information Systems and Technology (CIST) and at ICIS 2026.

Develops a biodynamics-guided deep learning framework for smartphone-based depression screening that quantifies real-world measurement noise through deep-ensemble uncertainty estimation, addressing the reliability problem that limits consumer-grade digital biomarkers.

Role: lead author; independently conceived and developed the biodynamics-guided deep learning framework.

PUBLICATIONS

Kim, J., Yoon, S., Chung, S., & Oh, W. (2025). Working Daily, Paid Monthly? Effects of On-Demand Wage Access on the Financial Engagement of Low-Wage Workers.

Information Systems Research 37(3):1463–1484. <https://doi.org/10.1287/isre.2023.0673>

Role: responsible for the causal machine learning analyses (double/debiased machine learning, causal forest); conducted the semistructured user interviews.

EDUCATION

Ph.D. in Management Engineering (Information Systems Track)

KAIST College of Business, Seoul, Korea | Mar. 2022 – Expected Aug. 2027

Advisor: Professor Wonseok Oh

Dissertation proposal defense completed, May 2026; final dissertation defense planned for April 2027

M.S. in Business Administration (Marketing)

Korea University Business School, Seoul, Korea | Mar. 2017 – Aug. 2019

Thesis: “Exploring Mechanism of Marriage Decision: Hierarchical Bayesian Approach”

Awards: Best Thesis Proposal Award; Best English Thesis Award (Korea University Business School Graduate School)

B.B.A. in Business Administration

Korea University, Seoul, Korea | Mar. 2009 – Feb. 2017

DISSERTATION

“When AI Enters the Healthcare Information Environment: Sensing, Treatment, and Markets”

Committee Chair: Professor Wonseok Oh

The dissertation examines what happens when AI relaxes long-standing information constraints in healthcare, across three essays spanning physiological screening, behavioral treatment delivery, and risk-based market design. It argues that the value of AI-generated health information is not fixed, but depends on the reliability of the information the AI produces, the mechanism through which it reaches individuals, and the market structure it enters.

- **Essay 1 (Job Market Paper):** “Screening Mental Health with Smartphones: Understanding Real-World Noise in Smartphone-Generated Physiological Data”: develops a biodynamics-guided deep learning framework for smartphone-based depression screening. *Under review, MIS Quarterly.*
- **Essay 2:** “Engaging for Better Sleep: User Interaction and Behavioral Change in a Mobile App-Based Digital Therapeutic for Insomnia”: randomized controlled trial (RCT)-based evaluation of a cognitive behavioral therapy for insomnia (CBT-I) digital therapeutic. *Presented at CIST 2025; in preparation for journal submission.*
- **Essay 3:** “AI Underwriting in Health Insurance, from Model to Market”: develops an AI underwriting model and examines how insurers should allocate decisions among rules, human judgment, and AI, in collaboration with KB Life Insurance, a major South Korean life insurer. *Field experiment in preparation on how the model’s risk information shapes underwriting decisions and individuals’ responses.*

WORKING PAPERS

- Kim, K., **Yoon, S.**, Kwon, H.E., & Oh, W. Rational App Curation: A Portfolio-Theoretic View of Mobile App Adoption and Churn. Preparing submission to *Information Systems Research*.
Role: co-developing the framework that connects the multiple discrete-continuous extreme value (MDCEV) model to a deep learning forecasting component, feeding MDCEV-estimated parameters into the network as inputs.
- Kim, K., **Yoon, S.**, Park, J., Lee, G., & Lee, D. What Algorithms Leave Behind: How Automatic Matching Reshapes Worker Capability in Ride-Hailing Platforms.
Accepted, ICIS 2026.
Targeted for Information Systems Research or Management Science.
Role: developing a structural model complementing the main empirical analysis.
- Park, J., **Yoon, S.**, Shin, D., & Cho, D. When Does a Pickup Become a Commitment? Dynamic Decision Boundaries and Temporal Graph Learning in Cashierless Retail Stores.
Preparing submission to Manufacturing & Service Operations Management.
Role: independently developed the deep learning model for the research question and context.

- Park, D., Yu, W., & **Yoon, S.** (authors in alphabetical order). Designing a Trustworthy AI Artifact for Emotion Measurement in Consumer Reviews: An Appraisal Theory-Driven Concept-Bottleneck Approach.

Preparing submission to Journal of Marketing.

Role: selected the theoretical framework (the OCC appraisal model) and developed a new prediction model.

- Bhaek, J., Choi, K., & **Yoon, S.** (authors in alphabetical order). Structure-aware Human–AI Reciprocal Estimation: Recovering Preference Structure from Generative AI under Human-Data Scarcity.

Under review, Information Systems Research.

Role: equal contribution; co-developed the estimator that pools response distributions from multiple generative AI services with scarce human-response data.

CONFERENCE PRESENTATIONS

- Kim, K., **Yoon, S.**, Park, J., Lee, G., & Lee, D. What Algorithms Leave Behind: How Automatic Matching Reshapes Worker Capability in Ride-Hailing Platforms. *ICIS 2026 (forthcoming)*.
- **Yoon, S.**, Oh, J., & Oh, W. Screening Mental Health with Smartphones: Understanding Real-World Noise in Smartphone-Generated Physiological Data. *CIST 2026 and ICIS 2026 (forthcoming)*.
- Kim, K., Kang, H., Lee, D., Lee, G., Park, J., & **Yoon, S.** Unintended Consequences of Algorithmic Management: Evidence from Automatic Matching Systems in Ride-Hailing Platforms. *PACIS 2026*.
- Kang, H., Kim, K., **Yoon, S.**, Lee, G., & Lee, D. Non-Punitive Governance in Algorithmic Markets: The Role of Tag-Based Feedback in Ride-Hailing. *WISE 2025*.
- **Yoon, S.**, Ko, S.G., & Oh, W. Engaging for Better Sleep: User Interaction and Behavioral Change in a Mobile App-Based Digital Therapeutic for Insomnia. *CIST 2025*.
- Kim, J., & **Yoon, S.** Could Self-Order Kiosks Drive Unequal Demand Trends? An Analysis of How Kiosks Influence Stores and Demand Variety. *WISE 2024*.
- Heo, W., **Yoon, S.**, Han, S.P., & Oh, W. When the Human–Algorithm Voice Connection Fails: Effects of Attribution Responses on User Engagement with AI-Enabled Smart Speakers. *CIST 2023*.
- Kim, J., **Yoon, S.**, & Chung, S. Working Daily, Paid Monthly? Effects of On-Demand Earned Wage Access on the Financial Well-Being of Low-Wage Workers. *CIST 2023*.

INDUSTRY & DATA PARTNERSHIPS

KB Life Insurance, a South Korean life insurer under KB Financial Group: Development of an AI underwriting model and a planned field experiment on AI-generated health risk information, Apr. 2026 – Present.

Kakao Mobility, a South Korean mobility-platform company: Data value assessment for platform-generated matching data, informing the What Algorithms Leave Behind working paper with approximately 48 million driver–passenger match attempts, Nov. 2023 – Mar. 2024.

Shinhan Card, a South Korean credit-card and financial-services company: AI-based call center service improvement, Aug. 2023 – Dec. 2024.

TEACHING EXPERIENCE

Instructor of Record

- Big Data Programming I, Department of Big Data Applications, Kyung Hee University, Mar. 2026 – Jul. 2026. Introductory Python programming course; sole instructor, 87 students. Teaching Evaluation: 94.67/100 (department average: 94.49/100)

Corporate & Executive Education (Instructor)

- AI-Assisted Software Development (“Vibe Coding” and Prompt Engineering) and Deep Learning. Kia Corporation, an automaker under Hyundai Motor Group, 2026.
- Machine Learning and Deep Learning for the Insurance Industry. Heungkuk Life Insurance, a Korean life insurer, 2025.
- Causal Machine Learning. TMAP Mobility, South Korea’s leading navigation and mobility platform, 2023.
- Machine Learning and Deep Learning. Samsung Electro-Mechanics, an electronic-components affiliate of Samsung Group, 2022.
- Machine Learning. Hyundai Motor Group, 2021.
- Machine Learning. LG Group, 2021.

Guest Lecturer

- Invited lecture in English on large language models (LLMs). Kyung Hee University, Fall 2025.
- Invited lecture in English on agentic AI. Kyung Hee University, Spring 2025.
- Digital and Platform Business (MBA), KAIST College of Business, Fall 2023. Topic: Difference-in-Differences Methodology.
- Korean Society & Culture (International MBA), KAIST College of Business, Fall 2022. Topic: Korea’s Economic Development (delivered in English while serving as the course’s teaching assistant).

Teaching Assistant

- AI-Driven Business Evolution (MBA), KAIST College of Business, Mar. 2026 – Jul. 2026. Worked with the faculty instructor to co-design the syllabus and evaluation rubrics; advised student teams building minimum viable product (MVP) web or mobile services with agentic AI or LLMs; reviewed business viability and provided feedback.
- Cloud Computing and Unstructured Data Analytics, KAIST College of Business, Mar. 2023 – Jul. 2023. Supported student projects on AWS-based collection and analysis of unstructured (text) data.

Curriculum Development

- Designed and taught the KCB Math Camp and KCB AI Camp (KCB: KAIST College of Business) for incoming master’s and doctoral students (curriculum, syllabus, and instruction), KAIST College of Business, Feb. 2024.
- Designed and recorded *Deep Learning for Computer Vision*, a self-paced online course (11 lectures, 13 h 20 min) for non-specialists with basic Python experience, Modulabs online learning platform, 2023. <https://class.modulabs.co.kr/classes/74>

SELECTED DOCTORAL COURSEWORK

Panel Data Econometrics; Microeconomic Analysis; Industrial Organization; Behavioral Economics; Quantitative Models for Marketing Decisions; Multivariate Statistical Analysis; Machine Learning for AI; Bayesian Machine Learning

GRANTS & AWARDS

- Selected Participant, INFORMS Information Systems Society (ISS) Doctoral Consortium, 2026.
- Doctoral Student Research Encouragement Grant, National Research Foundation of Korea (NRF), 2024–2026. KRW 20,000,000 (approx. USD 14,500) over two years.

- Best Thesis Proposal Award, Korea University Business School Graduate School, 2019.
- Best English Thesis Award, Korea University Business School Graduate School, 2019.

AD HOC REVIEWER

Conferences: ICIS (2022, 2023, 2024, 2026); PACIS (2023, 2024); CIST (2026)

Journals: Asia Pacific Journal of Information Systems (APJIS), 4 manuscripts; Decision Sciences Journal (DSJ), 1 manuscript

TECHNICAL SKILLS

Programming & Statistical Software: Python, R, Stata

Languages: Korean (native), English (fluent)